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The Editors-in-Chief
Machine Learning
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Dear Editors,

I am pleased to submit the manuscript **"Corpus-Specific Embedding-Model Selection for Retrieval-Augmented Generation: A Local Benchmarking Pipeline with Query-Level Uncertainty"** for consideration in *Machine Learning*.

Practitioners who build retrieval-augmented generation systems usually choose an embedding model from a public leaderboard. Those leaderboards measure behavior on public tasks, not on the practitioner's own documents, and they offer no way to tell whether a ranking gap observed on a local corpus is large enough to act on. The paper argues that this decision should be made per corpus, and it contributes the measurement instrument that makes doing so practical: a configurable pipeline that turns a folder of documents into a complete benchmark—chunks, synthetic queries, LLM-constructed relevance judgments, multi-model retrieval runs, query-level uncertainty estimates, and automated per-dataset and aggregate reports—without requiring any labeled data from the practitioner.

Three aspects may be of particular interest to the journal. First, the relevance judgments are pooled model-independently across the whole evaluated roster, with explicit negative labels retained, so the judged set is not biased toward any single candidate. Second, every corpus ranking carries paired bootstrap intervals, which turn out to be decisive rather than decorative: on the English education corpus the interval between the two leading candidates includes zero, so that corpus does not resolve them on retrieval quality and the choice falls to latency, whereas on both Turkish corpora the corresponding intervals exclude zero. Third, pool incompleteness is measured directly—9 of 150 seed chunks fall outside the pooled candidate set—rather than acknowledged in the abstract and then set aside.

The demonstration uses one frozen snapshot across three language- and domain-differentiated corpora: 60 documents, 1,914 chunks, five sentence-transformer candidates, 150 synthetic queries, and 9,296 judged query–chunk pairs including 952 positives. All intermediate artifacts are released, and every number stated in the paper is checked automatically against those artifacts. The paper is equally explicit about what the evidence does not establish: the queries are synthetic, the relevance judgments are LLM-generated without human validation, each language and domain condition is represented by a single corpus, and the retrieval geometry and chunking are fixed demonstration settings rather than tuned optima.

I should disclose that a preliminary version of this work was presented at the International Conference on Computer Science and Intelligence Systems (ICCSIS 2025) in Madrid, 10–11 August 2025, and was subsequently published by the conference organizers in the *International Journal of Advances in Electronics and Computer Science* 12(9). That version established feasibility on a single English corpus. The present manuscript extends it substantially: three language- and domain-differentiated corpora rather than one, model-independent pooled qrels, query-level bootstrap uncertainty, dataset-level reports as the primary selection artifact, and operational-cost evidence. The relationship is stated in Section 1 and in full in Section 2.6, the prior paper is cited, and a copy accompanies this submission.

The manuscript is not under review at any other journal or conference, and no overlapping submission is in progress. It has been prepared with the Springer Nature LaTeX template; the source files, style files, figures, and the compiled PDF are included.

Thank you for your consideration.

Yours sincerely,

Murat Karakaya